



INSTALLATION GUIDE

1. Introduction

This installation guide documents the deployment and configuration of an Ubuntu Server virtual machine using Oracle VirtualBox. It provides the required procedures for installing Ubuntu Server, configuring the server, verifying its functionality, and enabling SSH for secure remote administration.

2. Objectives

The objectives of this installation are to:

- Install Ubuntu Server in a virtualized environment.
- Configure the server according to the required specifications.
- Assign the required hostname.
- Configure a non-root administrative user.
- Verify network connectivity.
- Update the server.
- Install and verify OpenSSH Server.
- Document the installation process for future administrators.

The module states that the activity is intended to deploy a clean Ubuntu Server VM and verify its readiness for deployment.

3. System Requirements

Required Software

Software	Requirement
Virtualization Software	Oracle VirtualBox or VMWARE Workstation
Server Operating System	Ubuntu Server LTS
Installation Media	Ubuntu Server LTS ISO
Network	Internet connection

4. Virtual Machine Specifications

Create the Ubuntu Server VM using the following configuration:

Component	Configuration
Vm Name	Ubuntu-Server-Week03
RAM	4 GB
CPU	2 Virtual Processors
Storage	40 GB

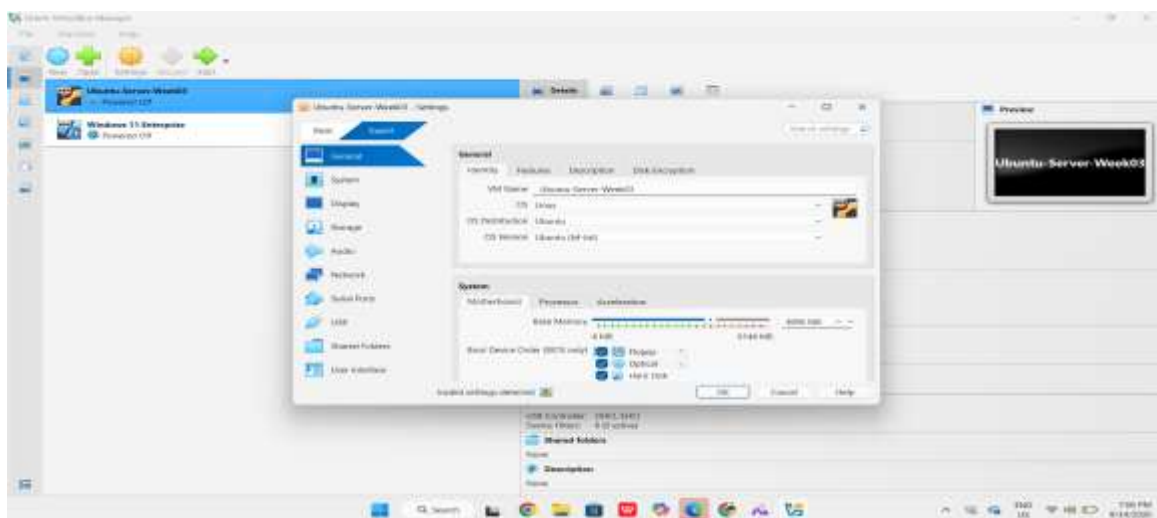


Disk Type	VDI/VMDK
Network	NAT
Optical Drive	Ubuntu Server ISO

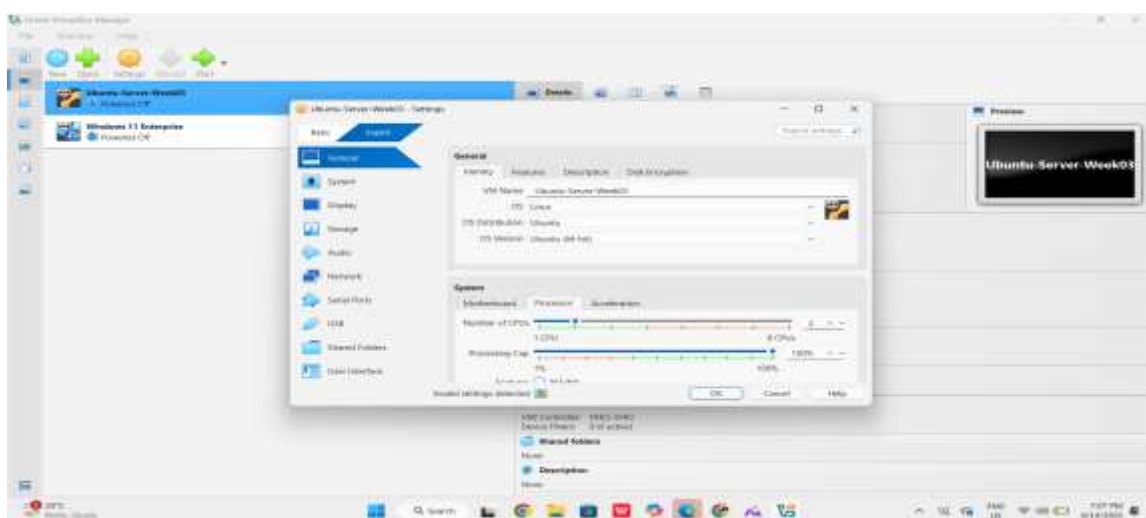
5. Ubuntu Server Installation

Step 1 Create the Virtual Machine

1. Open Oracle VirtualBox.
2. Select New.
3. Enter the VM name:



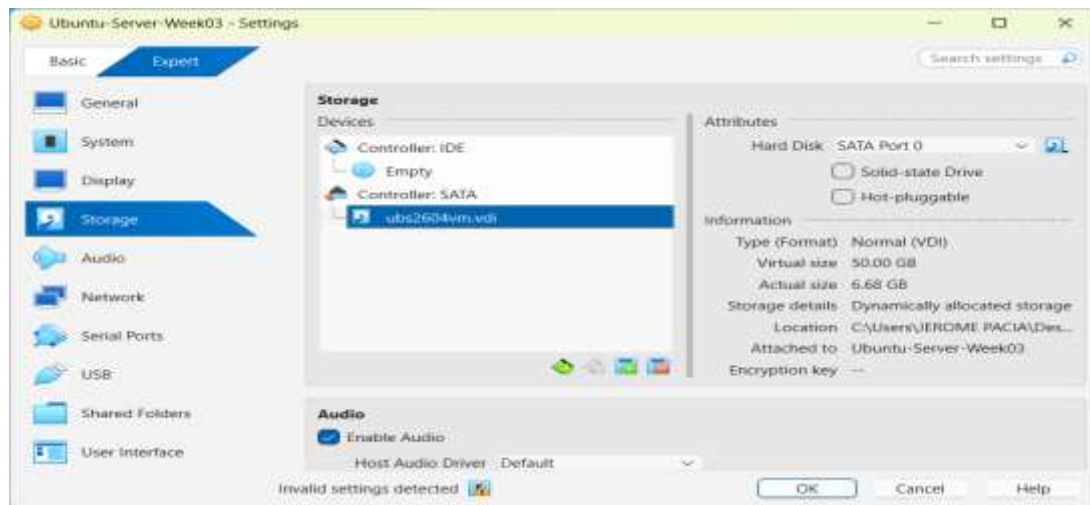
4. Select the Ubuntu Server ISO.
5. Allocate 4 GB RAM.
6. Assign 2 virtual processors.



7. Create a 40 GB virtual disk.



8. Configure the network adapter as NAT.



9. Start the virtual machine.

Step 2 Select Language

When the Ubuntu Server installer appears:

1. Select English.
2. Continue to the next screen.

Step 3 Configure Keyboard Layout

1. Select the appropriate keyboard layout.
2. Use **English (US)** if applicable.
3. Continue.

Step 4 Configure Network

1. Allow Ubuntu Server to detect the network.
2. Use **DHCP** unless instructed otherwise.
3. Record the assigned IP address.

Step 5 — Configure the Hostname

Set the hostname to: Server01

Step 6 Create the Administrative User

Create a non-root administrative user.

Use:

- **Full Name:** Your Name
- **Username:** yourlastname
- **Password:** Strong Password



Step 7 Configure Storage

Select:

Guided – Use Entire Disk

Document the following:

- Partition scheme
- File system
- Disk size

Step 8 Enable OpenSSH Server

When the installer reaches the SSH configuration section:

Select:

- **Install OpenSSH Server**

Do not install additional packages unless specifically instructed.

Step 9 Complete the Installation

1. Allow Ubuntu Server to finish installing.
2. Restart the virtual machine.
3. Remove the installation ISO if prompted.
4. Log in using the administrative account created during installation.

The module requires the VM to be restarted and the installation ISO removed if prompted.

6. Server Verification

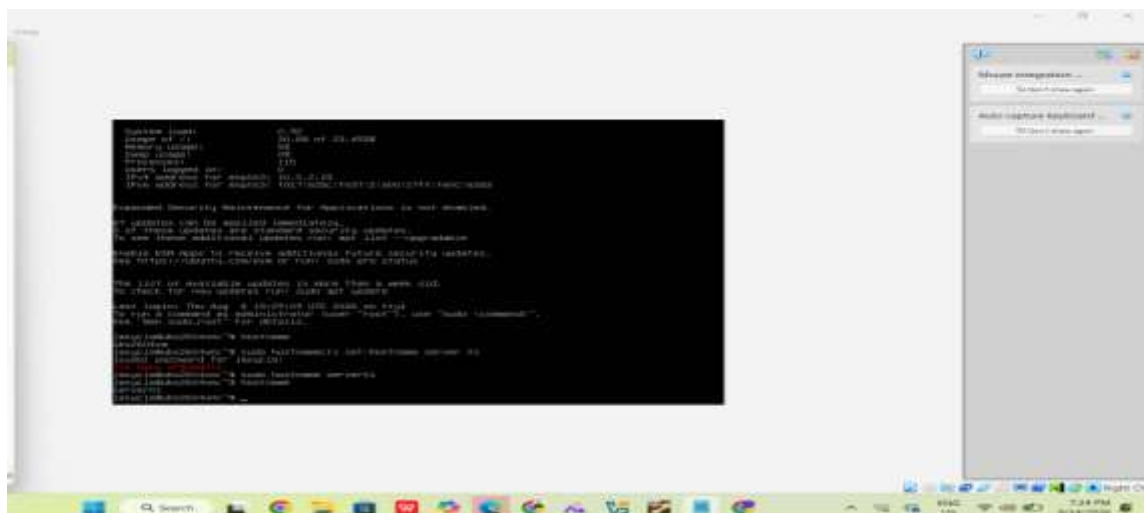
After logging into Ubuntu Server, perform the following verification tasks.

Run:

Hostname

Expected result:

server01





6.2 Verify IP Address

Run:

ip addr

```
ens33: flags=4096<UP,BROADCAST,MULTICAST> mtu 1500
    link/ether 08:00:27:14:78:0e brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.101 netmask 255.255.255.0 scope global dynamic ens33
        valid_lft 86400sec preferred_lft 14400sec
    inet6 fe80::208:1fff:fe14:780e scope link
        valid_lft forever preferred_lft forever
```

Identify and record the assigned IP address.

6.3 Test Internet Connectivity

Run:

ping -c 4 google.com

```
--- google.com ping statistics ---
 4 packets transmitted, 4 received, 0% packet loss, time 30.0ms
rtt min/avg/max/mdev = 0.428/0.512/0.584/0.076 ms
```

Verify that the server receives successful responses.



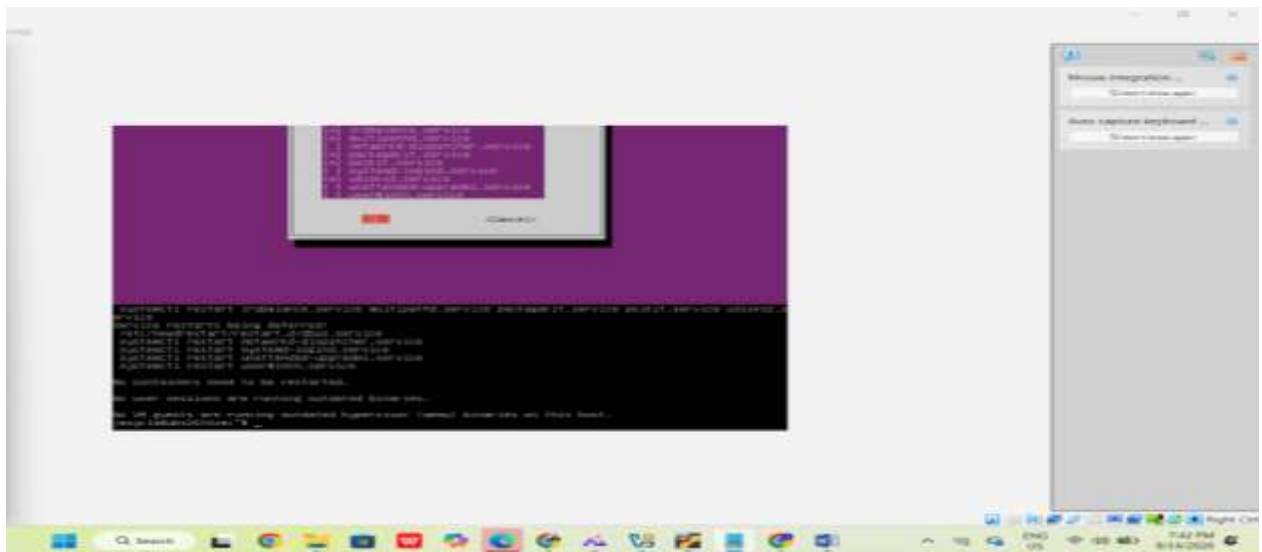
6.4 Update the Server

Run:

```
sudo apt update
```

Then:

```
sudo apt upgrade -y
```



Wait for the update process to complete

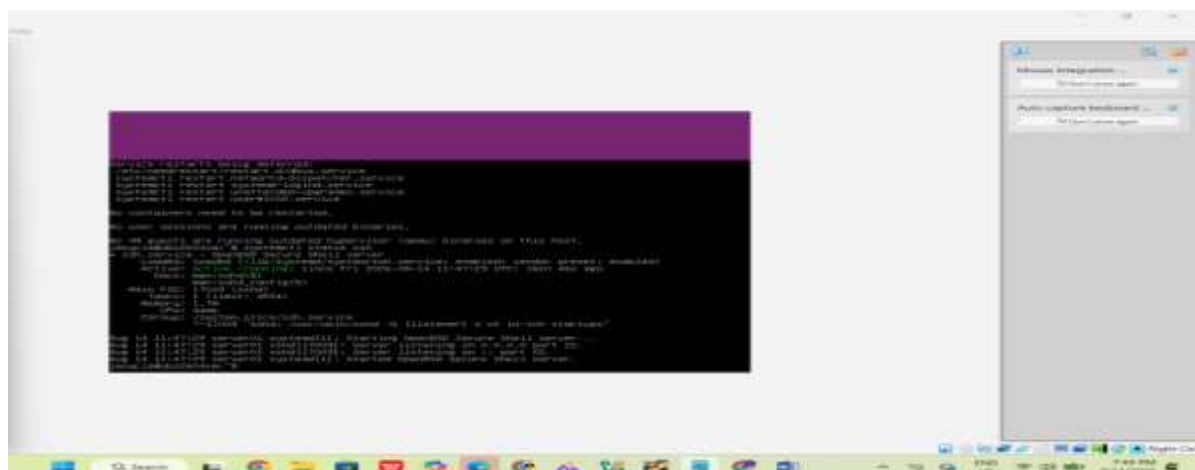
6.5 Verify SSH

Run:

```
systemctl status ssh
```

The SSH service should display:

Active: active (running)





7. Installation Verification Checklist

Requirement	Status
Ubuntu Server installed	✓
VM configured with required specifications	✓
Hostname configured	✓
Administrative user created	✓
Network configured	✓
IP address verified	✓
Internet connectivity verified	✓
Server updated	✓
OpenSSH Server installed	✓
SSH service verified as running	✓

BIOS vs UEFI Comparison

Category	BIOS	UEFI
Definition	BIOS (Basic Input/Output System) is standard firmware that prepares computer hardware and initiates the boot sequence.	A modern firmware interface called UEFI (Unified Extensible Firmware Interface) was created to take the place of traditional BIOS and offer a more adaptable and flexible boot environment.
Boot Process	BIOS loads the boot code from the device's boot sector after performing hardware initialization and looking for a bootable device.	UEFI initializes hardware and locates and launches an operating-system bootloader from the EFI System Partition using firmware boot managers.
Maximum Disk Support	MBR, which has a practical maximum boot-disk size of roughly 2 TB, is frequently used in traditional BIOS systems.	GPT, which supports disks much larger than 2 TB, is frequently compatible with UEFI.
Partition Style	Primarily connected to the partitioning of MBRs (Master Boot Records).	Primarily connected to partitioning using GPT (GUID Partition Table).
Security Features	The built-in security features of traditional BIOS are limited.	Supports security features like Secure Boot, which aids in stopping the



		loading of unauthorized boot software.
Speed	Has a more straightforward and constrained firmware initialization procedure.	Can offer a more effective boot process on modern hardware, tho the actual boot time is dependent on the operating system, firmware configuration, and hardware.
Advantages	Easy to use, compatible with legacy operating systems, and widely supported on older hardware.	Large disks, GPT, Secure Boot, modern boot managers, and more sophisticated firmware features are all supported.
Disadvantages	Limited security features, outdated architecture, restricted disk/partition capabilities, and legacy constraints.	More complicated than a conventional BIOS, and it might not work with very old hardware or operating systems.
Modern Usage	Primarily utilized in Legacy/CSM compatibility modes or on older computers.	The standard firmware environment on most modern PCs and is preferred for modern operating systems.

WHY UEFI HAS LARGELY REPLACED BIOS IN MODERN COMPUTERS

BIOS and UEFI are firmware systems that manage a computer's hardware initialization and booting the operating system. Nonetheless, UEFI was created to address several of the constraints linked to conventional BIOS. The UEFI standard delivers a uniform firmware framework with features that are more compatible with contemporary computer hardware and operating systems. A significant benefit of UEFI is its compatibility with the GUID Partition Table (GPT). Conventional BIOS systems are often linked to the Master Boot Record (MBR) partitioning method, which imposes restrictions on disk capacity and the count of primary partitions. GPT enables much larger storage devices and offers a contemporary partition structure. This renders UEFI especially beneficial as storage sizes have grown.

UEFI additionally offers a more adaptable boot procedure. Rather than depending on the conventional BIOS approach to find boot code in the disk's boot sector, UEFI can utilize firmware boot managers to identify operating-system bootloaders saved on an EFI System Partition. This offers a more organized method for starting modern operating systems. Another significant benefit is security. UEFI enables Secure Boot, a feature aimed at guaranteeing that only authorized boot software runs at startup. Secure Boot assists in safeguarding the initial phases of the

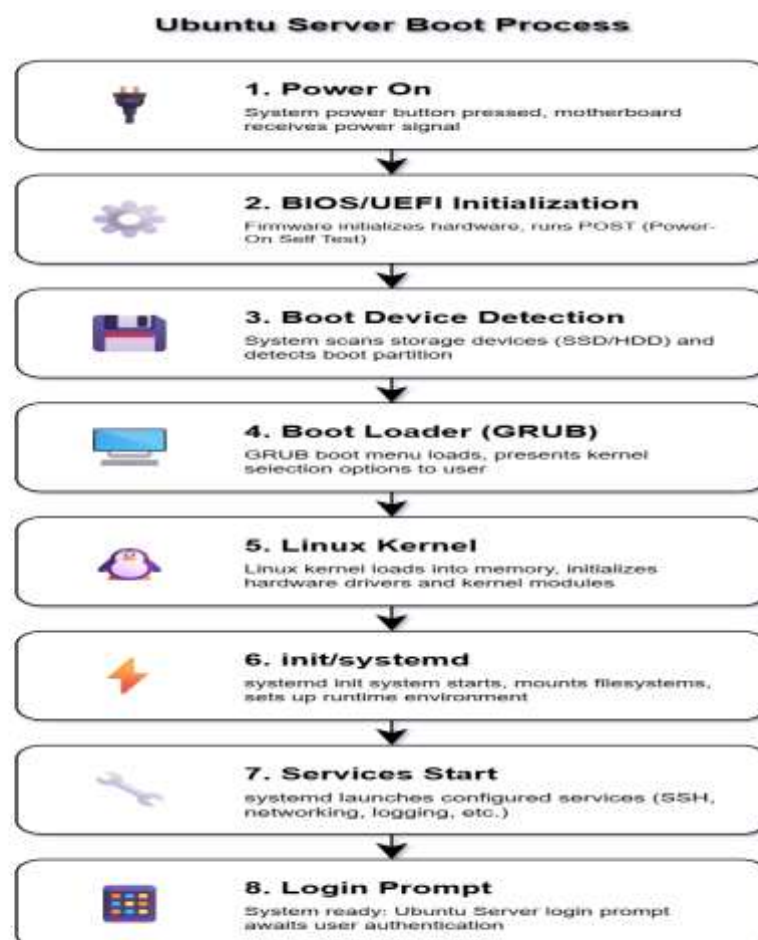


boot process against unauthorized or harmful software. Microsoft identifies Secure Boot as a crucial security element present in UEFI-based systems.

UEFI offers a more adaptable firmware environment compared to conventional BIOS. It can accommodate contemporary firmware applications, boot managers, drivers, and hardware setups. These features make UEFI more appropriate for contemporary computers compared to the relatively constrained BIOS setup. Another reason for UEFI's wide adoption is its compatibility with contemporary operating systems. Recent releases of Windows and Linux are built to support UEFI and GPT, and contemporary computer manufacturers are often delivering systems set up for UEFI as the standard configuration.

Although BIOS remains useful for older hardware and legacy operating systems, its limitations make it less suitable for today's computing environment. UEFI offers better support for large storage devices, modern partitioning, firmware security, and contemporary operating systems. For these reasons, UEFI has largely replaced traditional BIOS as the standard firmware interface on modern computers.

Boot Process Flowchart





Windows 11 Enterprise Installation

Windows 11 Enterprise was installed in a virtual machine using Oracle VirtualBox as the approved alternative to the Windows Server requirement. The installation used a Windows 11 Enterprise ISO and was configured as a separate virtual machine from the Ubuntu Server environment.

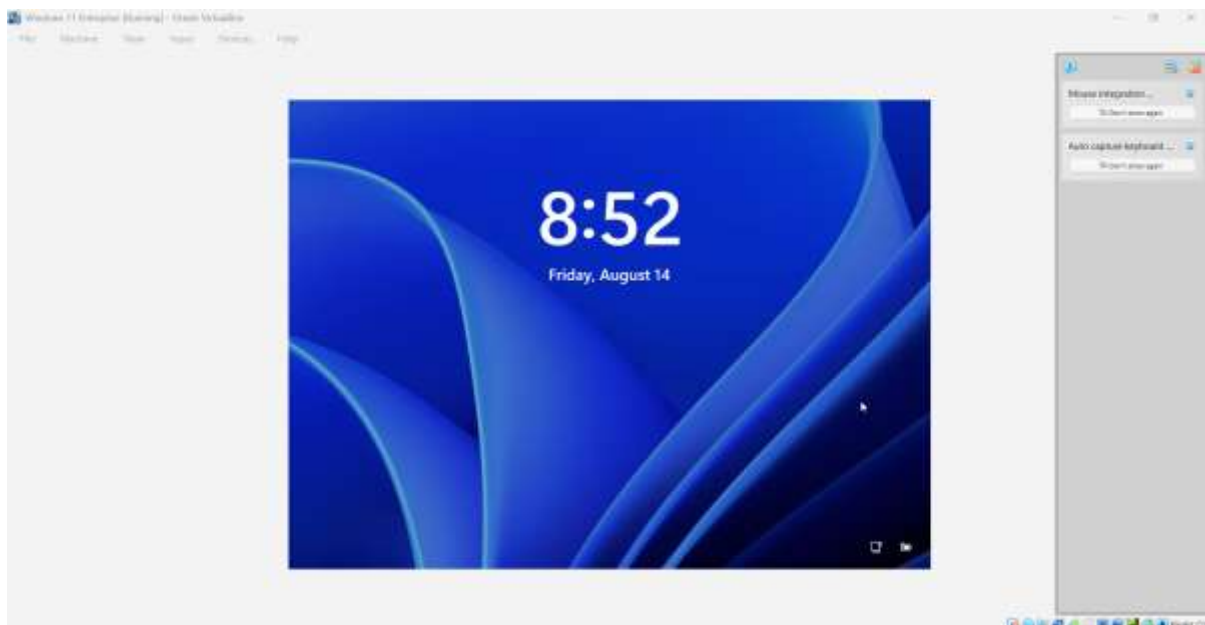
The installation process involved starting the virtual machine with the Windows 11 Enterprise ISO, following the Windows Setup wizard, selecting the appropriate installation options, and completing the operating system installation. After the installation was completed, the virtual machine was restarted and Windows 11 Enterprise successfully booted to the Windows desktop/login environment.

The installed Windows 11 Enterprise virtual machine was then used as the Windows operating system for the operating system comparison with Ubuntu Server and Rocky Linux.

Installation Configuration

Component	Configuration
Operating System	Windows 11 Enterprise
Virtualization Platform	Oracle VirtualBox
Installation Media	Windows 11 Enterprise ISO
Installation Type	Virtual Machine
Status	Successfully Installed

Installation Evidence:



Note: Windows 11 Enterprise was used as an approved substitute for the Windows Server requirement based on the instructor's approval.



OS Comparison

Windows 11 Enterprise vs Ubuntu Server vs Rocky Linux

Category	Windows 11 Enterprise	Ubuntu Server
Licensing	Proprietary Microsoft operating system	Open-source and free to use
User Interface	Graphical desktop interface	Primarily command-line based
Package Management	Microsoft Store, winget, and Windows installers	APT and .deb packages
Security	Microsoft Defender, Windows Firewall, BitLocker, Secure Boot, and enterprise security policies	Linux permissions, AppArmor, firewall tools, and SSH
Performance	Designed mainly for desktop and workstation workloads	Designed for server workloads and efficient resource usage
Typical Use Cases	Enterprise desktops, workstations, business applications, and development	Web hosting, databases, cloud services, file sharing, and network services
Advantages	User-friendly graphical interface, strong Microsoft ecosystem, and enterprise security features	Flexible, cost-effective, open-source, and widely supported
Disadvantages	Proprietary licensing and generally higher resource requirements	Requires Linux and command-line administration knowledge

Lessons Learned:

This task offered hands-on experience in setting up, configuring, and validating an Ubuntu Server within a virtualized setting. I discovered how to set up key server configurations like the hostname, keyboard layout, disk partition, network connection, and SSH service. I also discovered the significance of checking every configuration post-installation. Commands like `hostname`, `ip addr`, `ping`, `sudo apt update`, and `systemctl status ssh` verified that the server was correctly set up and functioning.

A significant takeaway was resolving command-line issues. Mistakes in command syntax can lead to failures in installation or configuration tasks, making it essential for system administrators to thoroughly read error messages and verify the proper command format. The BIOS and UEFI activity furthered my understanding of how firmware technologies influence the computer boot procedure. Developing the Ubuntu boot process flowchart enhanced my comprehension of the sequence from system startup, firmware setup, bootloader activation, kernel loading, system services, and ultimately the login prompt.



In summary, the experience enhanced my knowledge of virtualization, Linux server management, problem-solving, system validation, and technical writing. It also highlighted the necessity of precisely recording system configurations so another administrator can replicate and uphold the setup.

References:

UEFI Forum — UEFI Specifications

<https://uefi.org/specifications>

Microsoft Learn — UEFI Mode vs. Legacy BIOS Mode

<https://learn.microsoft.com/en-us/windows-hardware/manufacture/desktop/boot-to-uefi-mode-or-legacy-bios-mode>

Microsoft Learn — Secure Boot

<https://learn.microsoft.com/en-us/windows-hardware/design/device-experiences/oem-secure-boot>

Red Hat — Red Hat Enterprise Linux Documentation (UEFI/BIOS)

https://docs.redhat.com/en/documentation/red_hat_enterprise_linux/

Ubuntu Documentation — Ubuntu Official Documentation

<https://documentation.ubuntu.com/>